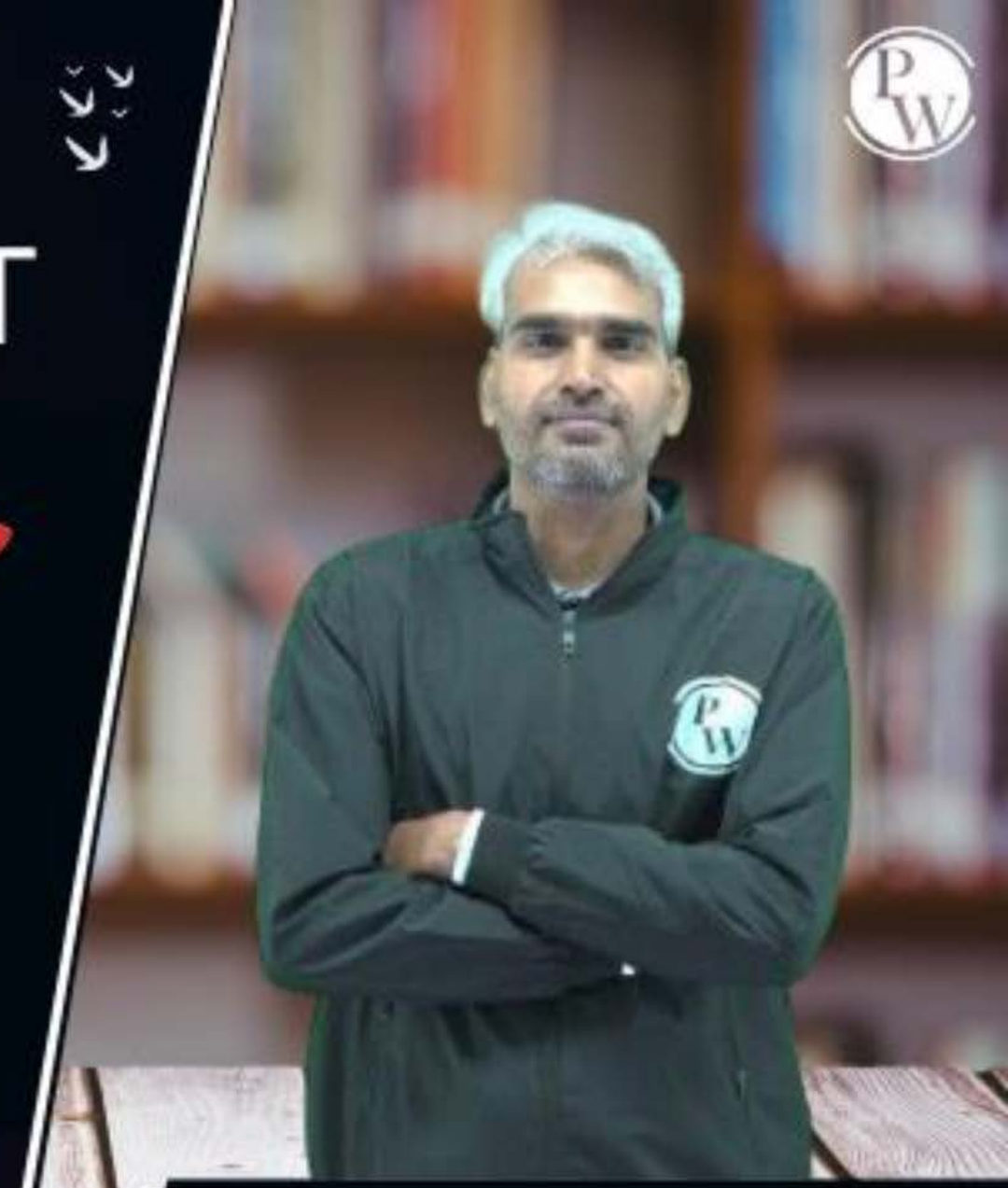
Computer Science & IT



Discrete Mathematics

Set Theory & Algebra

Lecture No. 14



By- Vishal Sir

Recap of Previous Lecture



Topic — Least upper bound (LUB)

Topic — Greatest lower bound (GLB)

Topic — Minimal and maximal elements in a POSET

Topic — Minimum (Least) element of POSET

Topic — Maximum (Greatest) element of POSET

Slide

Topics to be Covered



Topic Lattice

Topic Hasse diagram / POSET diagram

Slide

Note:-

In a POSET, lub and/or glb need not exist for every pair of elements of the set

Lattice



LUB

Topic : Join Semi Lattice

* A POSET in which least upper bound exists for every pair of elements of the set is called **Join - semi lattice**

{ glb may or may not exist for every pair of elements. }

Slide



Topic : Meet Semi Lattice

GLB

* A POSET in which greatest lower bound exists for every pair of elements of the set is called a **Meet semi lattice**.

{ lub may or may not exist for every pair of elements }

Slide



Topic : Lattice

A POSET in which both least upper bound (lub) as well as greatest lower bound (glb) exists for every pair of elements of the set is called a lattice

* A POSET that is a Join semi lattice as well as a Meet semi lattice is called a lattice

Slide

Eg ① $A_1 = \{2, 3, 4, 6, 15\}$ and $(A_1, \div)$ is a POSET (i) Check if lub exists for Every pair of elements (ii) Check if glb exists for Every pair of elements

② $A_2 = \{2, 3, 4, 6, 12\}$ and $(A_2, \div)$ is a POSET (i) Check if lub exists for Every pair of elements (ii) Check if glb exists for Every pair of elements

③ $A_3 = \{1, 2, 3, 4, 6, 15\}$ and $(A_3, \div)$ is a POSET (i) Check if lub exists for Every pair of elements (ii) Check if glb exists for Every pair of elements

④ $A_4 = \{1, 2, 3, 4, 6, 12\}$ and $(A_4, \div)$ is a POSET (i) Check if lub exists for Every pair of elements (ii) Check if glb exists for Every pair of elements

Eg ① $A_1 = \{2, 3, 4, 6, 15\}$ and $(A_1, \div)$ is a POSET

(i) Check if lub exists for Every pair of elements

(ii) Check if glb exists for Every pair of elements

lub (6, 15) = Does not exist

∴ Not a Join semi lattice

glb (2, 3) = Does not exist

∴ Not a Meet semi lattice

Eg ② $A_2 = \{2, 3, 4, 6, 12\}$ and $(A_2, \div)$ is a POSET

(i) Check if lub exists for Every pair of elements
(ii) Check if glb exists for Every pair of elements

① We can observe that lub exist for Every pair of elements of POSET, ∴ Given POSET is a Join semi lattice

② glb(2, 3) = Does not exist
└ ∴ Not a meet semi lattice

Eg ③ $A_3 = \{1, 2, 3, 4, 6, 15\}$ and $(A_3, \div)$ is a POSET (i) Check if lub exists for Every pair of elements (ii) Check if glb exists for Every pair of elements

① lub (6, 15) = Does not exist
↳ ∴ Not a join semi lattice

② '1' is present w.r.t. divides ($\div$)
∴ Glb will exist for Every pair of elements.
↳ Hence given POSET is a meet semi lattice

eg ④ $A_4 = \{1, 2, 3, 4, 6, 12\}$ and $(A_4, \div)$ is a POSET

(i) Check if lub exists for Every pair of elements

(ii) Check if glb exists for Every pair of elements

① Lub exist for Every pair of elements

∴ it is a join semi lattice

② Glb exist for Every pair of elements

∴ it is a meet semi lattice

Hence given POSET is a "Lattice"

Note: If 'n' is any positive integer, then 'D_n' is defined as a set of all positive divisors of 'n'.

i.e. $D_1 = \{1\}$

$D_2 = \{1, 2\}$

$D_3 = \{1, 3\}$

$D_4 = \{1, 2, 4\}$

$D_5 = \{1, 5\}$

$D_6 = \{1, 2, 3, 6\}$

$D_{12} = \{1, 2, 3, 4, 6, 12\}$

$D_{21} = \{1, 3, 7, 21\}$

* Note:-

① If D_n is the set of all positive divisors of a positive integer 'n', then

$(D_n, \div)$ is always a lattice.

② If 'A' is any finite set, and $P(A)$ is the Power set of set A, then

$(P(A), \subseteq)$ is always a lattice

③ If A is any set of real numbers, then

$(A, \leq)$ is always a lattice



Topic : Lattice

A lattice is denoted by $[L, \vee, \wedge]$

represent lub

represent glb

Name of lattice

it is actually the Name of the set in the POSET w.r.t. lattice

Slide

Properties w.r.t. Lattice

- Let $[L, \vee, \wedge]$ be a lattice, then following properties will always hold true in a lattice

1. Idempotent Law:
   (i) $a \vee a = a$, $\forall a \in L$
   (ii) $a \wedge a = a$, $\forall a \in L$
2. Commutative Law
   (i) $a \vee b = b \vee a$, $\forall a, b \in L$
   (ii) $a \wedge b = b \wedge a$, $\forall a, b \in L$
3. Associativity Law
   (i) $(a \vee b) \vee c = a \vee (b \vee c)$, $\forall a, b, c \in L$
   (ii) $(a \wedge b) \wedge c = a \wedge (b \wedge c)$, $\forall a, b, c \in L$
4. Absorption Law
   (i) $a \vee (a \wedge b) = a$, $\forall a, b \in L$
   (ii) $a \wedge (a \vee b) = a$, $\forall a, b \in L$

These four properties will always hold true in any lattice

Note:- Distributive Property need not hold true for all lattices.

i.e. (i) $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$, $\forall a, b, c \in L$

and (ii) $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$, $\forall a, b, c \in L$

If above distributive property holds true in a lattice, then that lattice is called a distributive lattice

If distributive property does not hold true for at least one combination of a, b & c, then lattice is not a distributive lattice

* Note :→

Important

In a lattice least upper bound as well as greatest lower bound must exist for every pair of elements and they must be unique

Hasse Diagram

Constructed for POSET

Topic : Hasse Diagram / POSET Diagram

In a Hasse diagram of a POSET, → Reflexive, Anti-symmetric, Transitive

1. There is a vertex corresponding to every element of set.
2. There is an edge from vertex a to vertex b if and only if a relates to b and there is no element x in the set such that a relates to x, and x relates to b. (Transitivity is implied in the Hasse diagram not represented explicitly)
3. No self-loop on the vertices (i.e. reflexivity is implied in the Hasse diagram not represented explicitly).
4. It is not directed but it uses implied upward orientation.

Slide



Topic : Hasse Diagram / POSET Diagram

Draw the hasse diagram for the following POSETs

1) $(\{-1,0,2.5,4,6\},\leq)$
2) $(D_6,\div)$
3) $(D_{12},\div)$
4) $(\{2,3,4,6\},\div)$
5) $(\{2,3,6,12\},\div)$
6) $(\{1,2,3,4,6,9\},\div)$

(1) $(\{-1, 0, 2.5, 4, 6\}, \leq)$ ← This POSET is also a TOSET

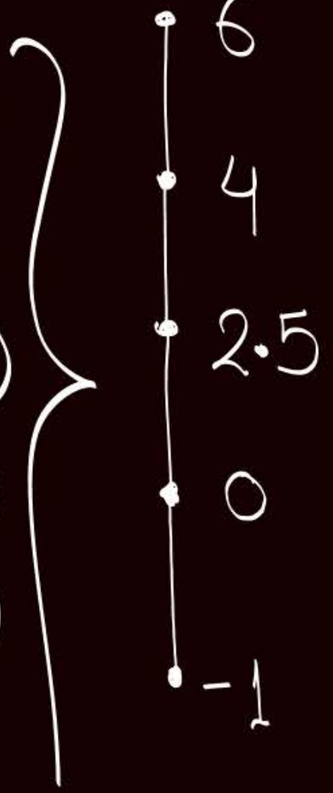
↓

Whenever we construct the Hasse diagram w.r.t. a TOSET it will look like a linear chain.

⇓

∴ TOSET are also known as "Linearly ordered set".

"$\leq$" = $\{(-1,-1), (-1,0), (-1,2.5), (-1,4), (-1,6)$
$(0,0), (0,2.5), (0,4), (0,6)$
$(2.5,2.5), (2.5,4), (2.5,6)$
$(4,4), (4,6)$
$(6,6)\}$



Q:- Let $A = \{a, b, c, d\}$

How many Total order relation Can be defined on set A.

i.e. TOSET

└ Hasse diagram w.r.t. TOSET will look like a linear chain.

i.e.,



'4' elements of the set can be placed at '4' vertices of Hasse diagram in 4! ways and each Hasse diagram will represent a different Total order relation

∴ 4! different Total order relation are possible

Note:- If A is a set with 'n' elements, then $n!$ different total order relation are possible on set A.

- Let $A = \{a, b, c, d\}$
and assume two different Hasse diagrams w.r.t. TOSETs (A, R_1) and (A, R_2) respectively

(i) Hasse diagram w.r.t. (A, R_1)



Find Rel^n R_1

$$= \{(a,a)\ (a,b)\ (a,c)\ (a,d)\ (b,b)\ (b,c)\ (b,d)\ (c,c)\ (c,d)\ (d,d)\}$$

(ii) Hasse diagram w.r.t. (A, R_2)



Find Rel^n R_2

$$= \{(b,b)\ (b,c)\ (b,d)\ (b,a)\ (c,c)\ (c,d)\ (c,a)\ (d,d)\ (d,a)\ (a,a)\}$$

Note:- If POSET are different, then Hasse diagram will also be different on same set

eg: Let $A = \{a, b, c, d\}$

(i) $R_1 = \{(a,a), (b,b), (c,c), (d,d)\}$

Partial order Relⁿ $\Rightarrow$ $\therefore (A, R_1)$ is a POSET

Hasse diagram is



If elements are not Comparable then they may be at same level

(ii) $R_2 = \{(a,a), (b,b), (c,c), (d,d), (a,b)\}$

$\therefore (A, R_2)$ is a POSET

Hasse diagram is,



(iii) $R_3 = \{(a,a), (b,b), (c,c), (d,d), (b,a)\}$

$\therefore (A, R_3)$ is a POSET

Hasse diagram is,

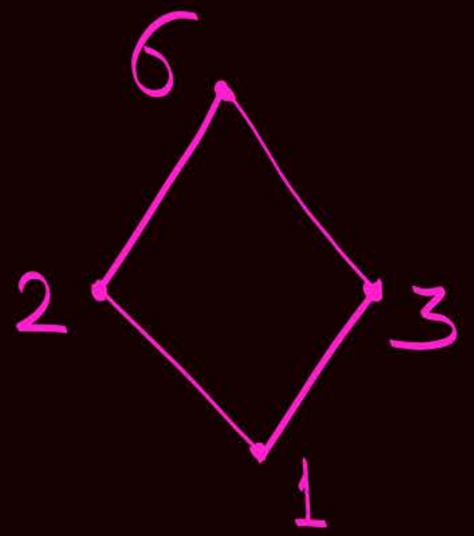


Q (ii) Construct the Hasse diagram for POSET,

$(D_6, \div)$

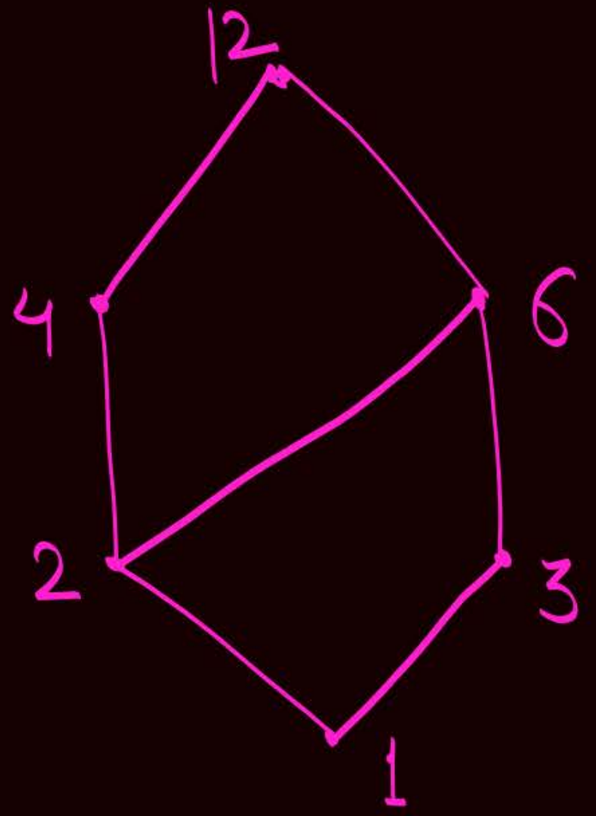
$D_6 = \{1, 2, 3, 6\}$

i.e, $(\{1, 2, 3, 6\}, \div)$

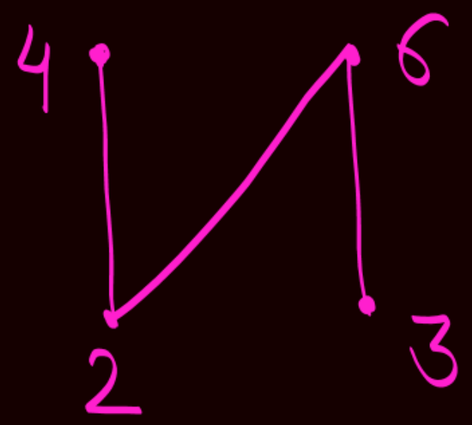


Q(iii) Construct the Hasse diagram for POSET, $(D_{12}, \div)$

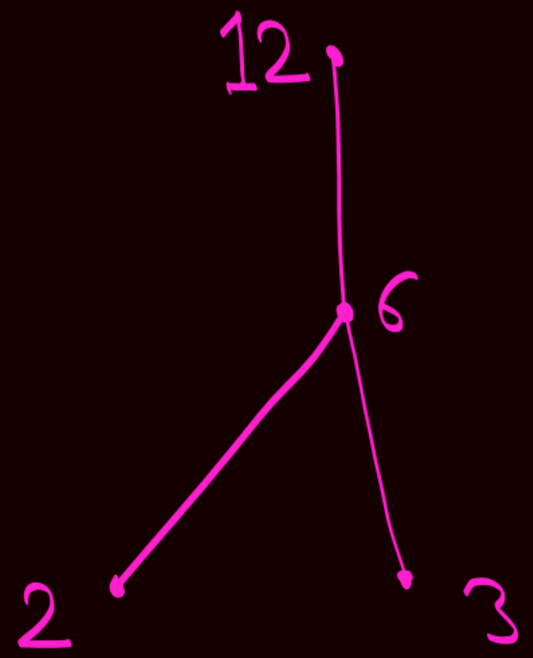
$D_{12} = \{1, 2, 3, 4, 6, 12\}$



Q (iv) Construct the Hasse diagram for POSET, $(\{2, 3, 4, 6\}, \div)$

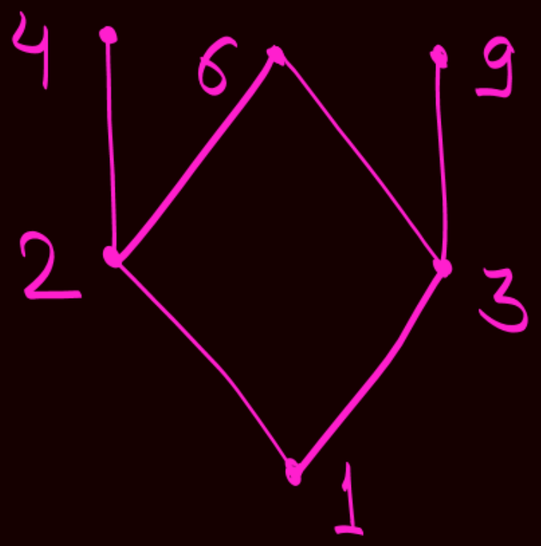


Q (V) Construct the Hasse diagram for POSET, $(\{2, 3, 6, 12\}, \div)$



Q (vi) Construct the Hasse diagram for POSET,

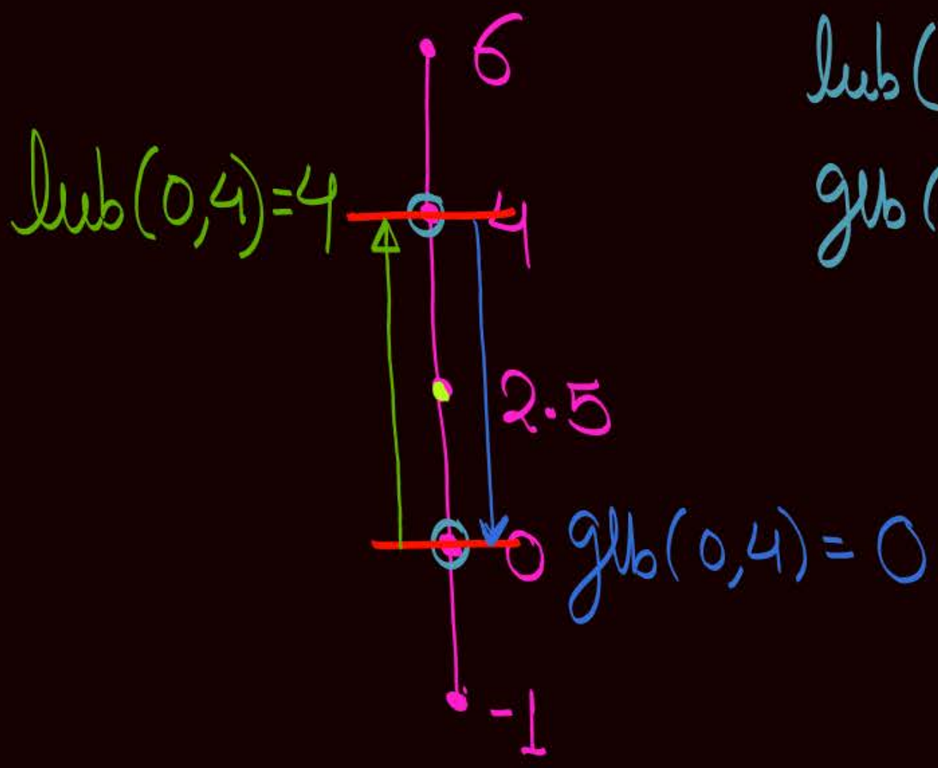
$(\{1, 2, 3, 4, 6, 9\}, \div)$



Understand lub & glb w.r.t. Hasse diagram

(1) $\left(\{-1, 0, 2.5, 4, 6\}, \leq\right)$ ← less than or equal

Hasse diagram for given POSET:



lub(0,4) = ?

glb(0,4) = ?

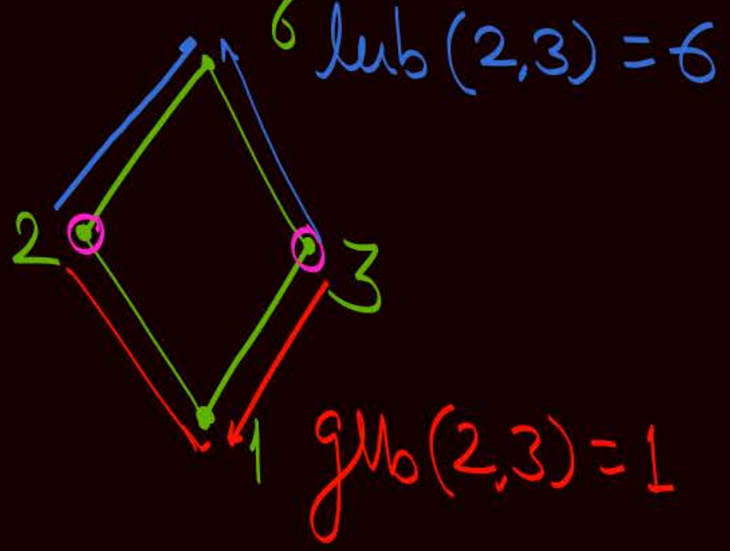
If we choose two elements on the same line, then the element at the upper side on the line is "lub". And the element at the lower side on the line is "glb"

- In the given POSET "lub" as well as "glb" exist for Every pair of elements and it is unique, Hence given POSET is a lattice

Note:-

Every TOSET is always a lattice but Converse of the Statement need not be true (i.e. Every lattice need not be a TOSET

Q (ii) Construct the Hasse diagram for POSET, $(D_6, \div)$

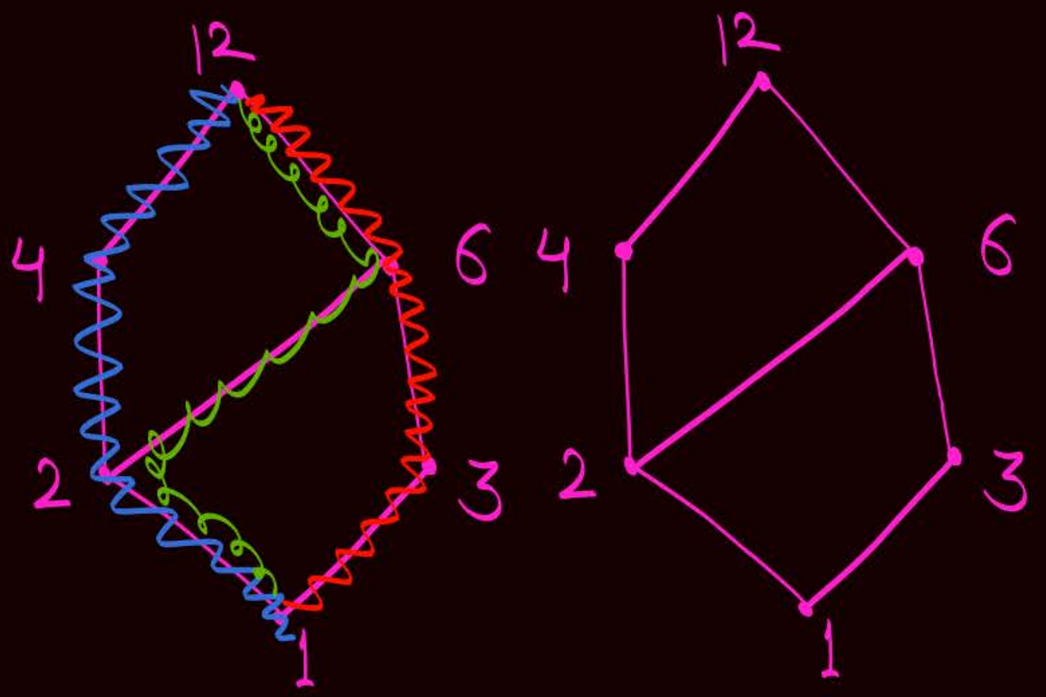


"Lub" as well as "glb" Exist for Every Pair of Elements.

∴ Given POSET is a lattice

Q(iii) Construct the Hasse diagram for POSET,

$(D_{12}, \div)$



lub (2, 3) = 6

glb (2, 3) = 1

lub (3, 4) = 12

glb (3, 4) = 1

lub (4, 6) = 12

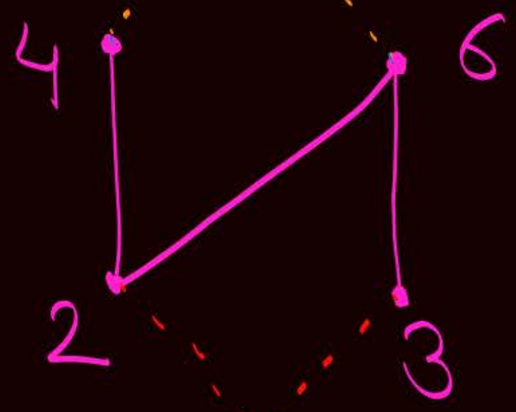
glb (4, 6) = 2

"Lub" as well as "Glb" exist for every Pair of elements of POSET

∴ Given POSET is a lattice

Q (iv) Construct the Hasse diagram for POSET,

$(\{2, 3, 4, 6\}, \div)$



lub(4,6) = Does not exist

∴ Not a Join semi lattice

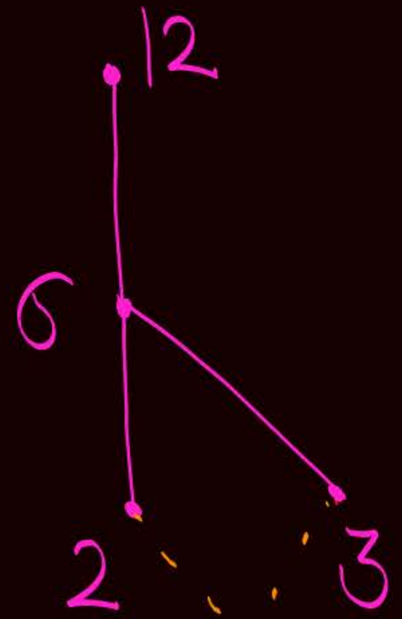
glb(2,3) = Does not exist

∴ Not a Meet semi lattice

Neither a join semi lattice nor a Meet semi lattice

Q (V) Construct the Hasse diagram for POSET,

$(\{2, 3, 6, 12\}, \div)$



glb(2,3): does not exist.

lub(2,3) = 6 :-

└ Lub exist for Every pair of elements

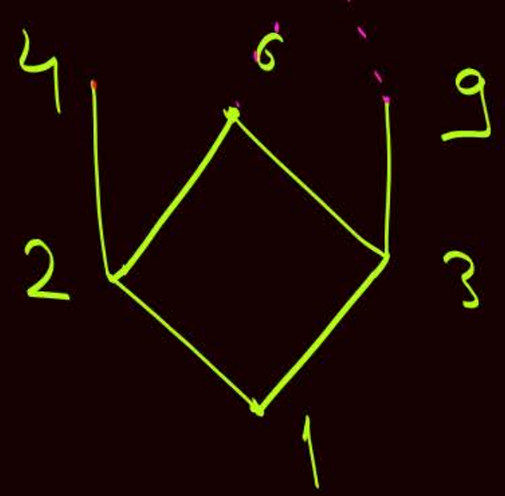
∴ It is a Join semi lattice

glb(2,3) = DNE

└ ∴ Not a meet semi lattice, and hence not a lattice

Q (vi) Construct the Hasse diagram for POSET,

$(\{1, 2, 3, 4, 6, 9\}, \div)$



lub(6, 9) = DNE

∴ Not a Join semi lattice

Hence, Not a lattice

Glb exist for Every pair of elements

∴ It is a meet semi lattice

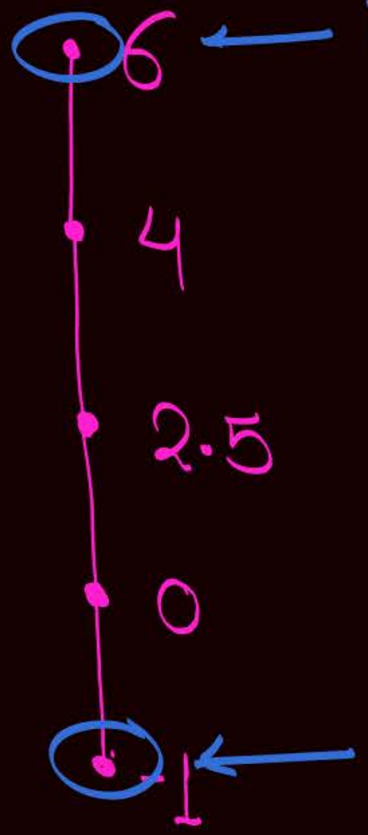
Maximal & Minimal Elements

and

Maximum & Minimum element

w.r.t. Hasse diagram

(1) $(\{-1, 0, 2.5, 4, 6\}, \leq)$



← Maximal element
(it is unique)
↳ ∴ Maximum as well

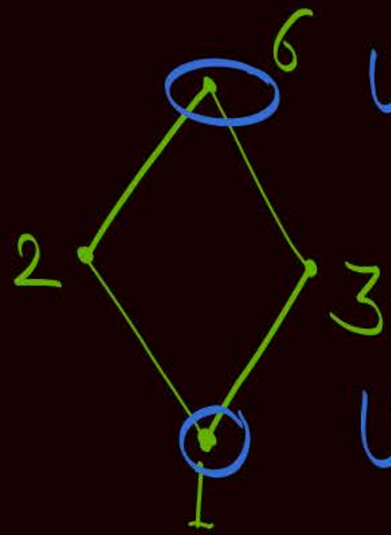
← Minimal element
(it is unique)
↳ ∴ Minimum as well

Q (ii) Construct the Hasse diagram for POSET,

$(D_6, \div)$

$D_6 = \{1, 2, 3, 6\}$

i.e, $(\{1, 2, 3, 6\}, \div)$



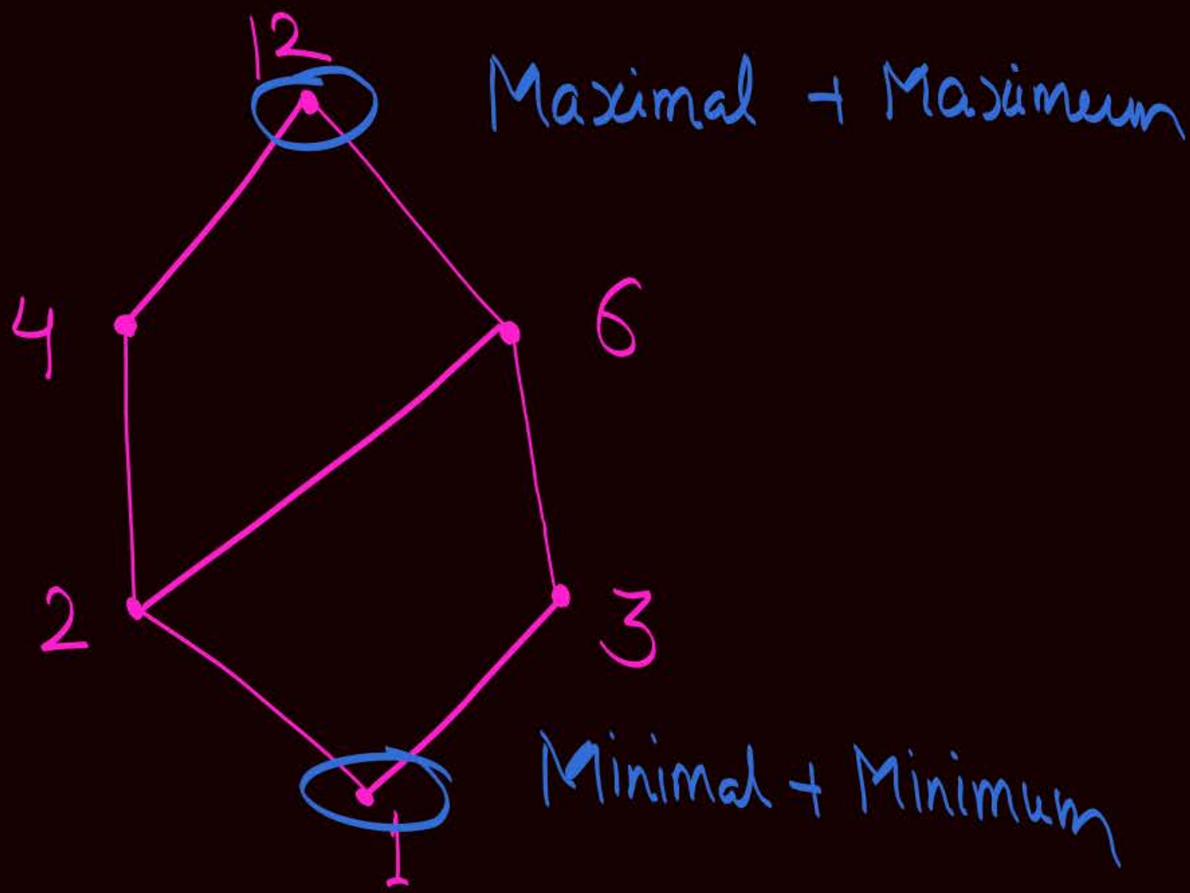
Unique maximal
∴ Maximum as well

Unique minimal
∴ Minimum as well

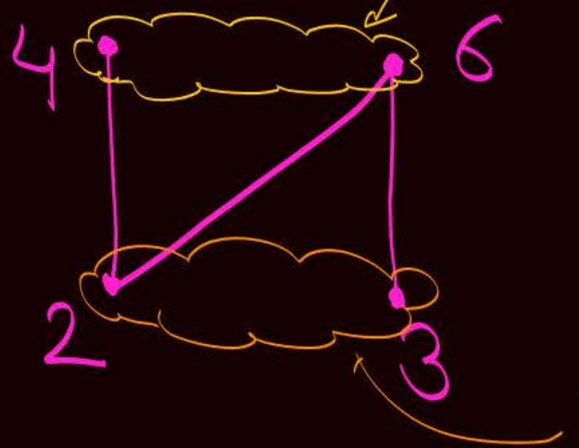
Q(iii) Construct the Hasse diagram for POSET,

$(D_{12}, \div)$

$D_{12} = \{1, 2, 3, 4, 6, 12\}$



Q (iv) Construct the Hasse diagram for POSET, $(\{2, 3, 4, 6\}, \div)$



Two Maximal
∴ No maximum

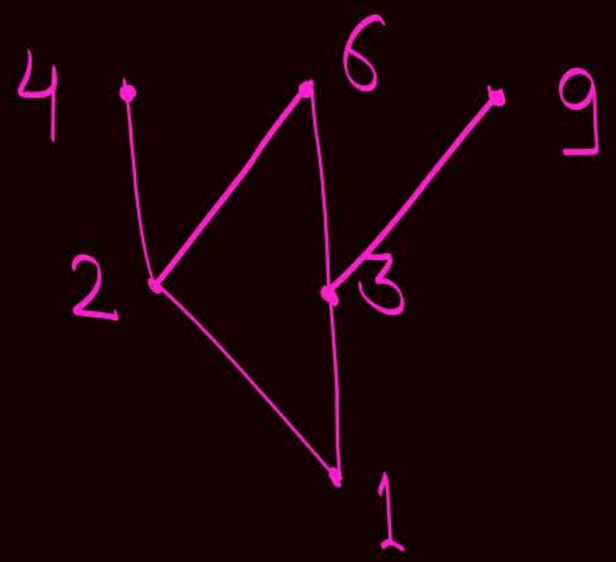
Two minimal
∴ No minimum

Q (V) Construct the Hasse diagram for POSET, $(\{2, 3, 6, 12\}, \div)$

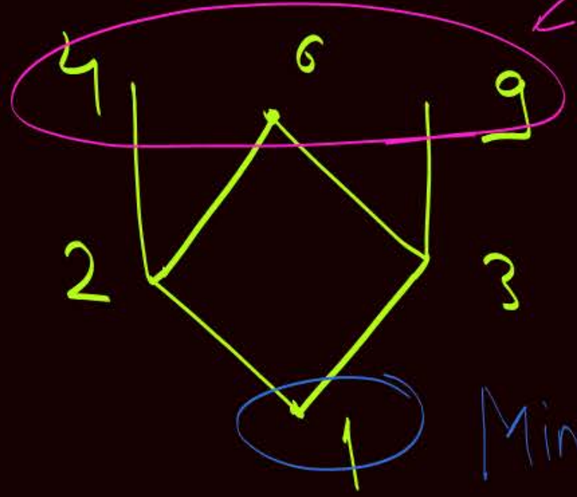


Q (vi) Construct the Hasse diagram for POSET,

$(\{1, 2, 3, 4, 6, 9\}, \div)$



$\equiv$



More than one maximal
∴ No maximum

Minimal + Minimum

Note :-

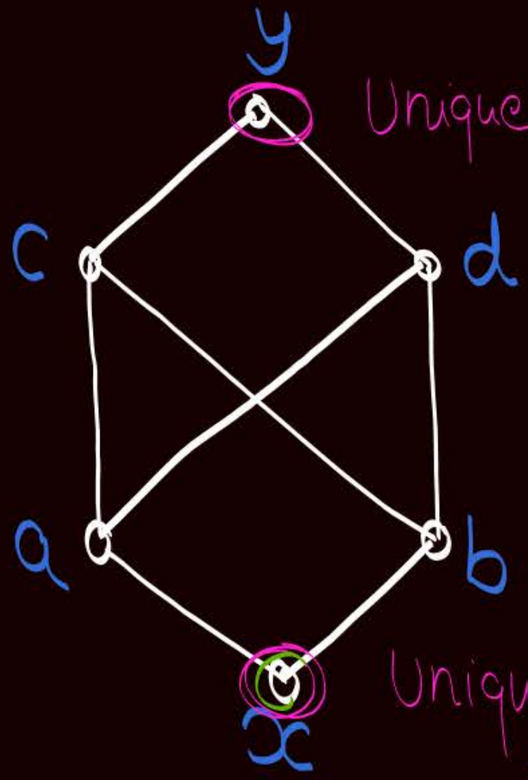
① In a POSET if there exist two or more maximal elements then it can not be a Join semi lattice {∴ Not a lattice}

lub will not exist for any two maximal elements

② In a POSET if there exist two or more minimal elements then it can not be a Meet semi lattice {∴ Not a lattice}

glb will not exist for any two minimal elements

eg:→ Consider the following POSET diagram
Check whether the POSET corresponding to given POSET diagram is a lattice or not?



Unique maximal
∴ Maximum

Unique minimal
∴ Minimum

* Both Maximum & Minimum elements exist in the POSET.

But (i) lub(a,b) = either we say that DNE
or
We say that both 'c' & 'd' are lub

Similarly (ii) glb(c,d) = either we say that DNE
or
We say that both 'a' & 'b' are glb

↳ In any case given POSET is not a lattice

→ There exists some POSET, in which both minimum as well as maximum element exist but still they are not lattices.

Note:→ It is not necessary for a lattice to have maximum and/or minimum element, because the set may be an infinite set

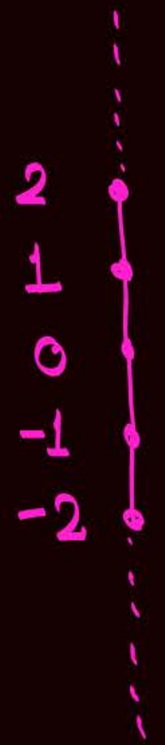
$(N, \leq)$ is a lattice,

Maximum element : Does not exist

Minimum element = 1

eg: $(Z, \leq)$ is a POSET

Hasse diagram



It is a TOSET

∴ lub exist for every Pair of elements as well as glb exist for every Pair of elements

Hence it is a lattice,

But, maximum element does not exist

And Minimum element also does not exist.

* Minimum and Maximum element can exist even if the set corresponding to the POSET is an infinite set

eg: $A = \{x \mid x \in R \text{ and } 0 \leq x \leq 1\}$

$(A, \leq)$ is a POSET

Where set is infinite, still both minimum as well as maximum element exist.

- If lattice is a finite lattice, then
  Both minimum as well as maximum elements
  will exist.

H.W. Q:- Let $A = \{1, 2, 3, 4, 5\}$

and let $R = \{(1,1), (2,2), (3,3), (4,4), (5,5), (1,2), (3,2), (4,5)\}$ is a Partial order on set A.

How many total order relation are possible on set A such that relation 'R' is included in each total order relation

H.W. Q:- Let $A = \{1, 2, 3, 4, 5\}$
How many partial order relation are possible on set A, such that the cardinality of each partial order relation is exactly '7'



2 mins Summary

Topic Lattice

Topic Hasse diagram / POSET diagram

Slide

THANK - YOU